# Lakshya JEE AIR (2025)

## WEEKLY TEST - 30

**DURATION : 180 Minutes** | **DATE : 25/11/2024** | **M.MARKS : 300**

### Topics Covered

| | |
|---|---|
| **Physics:** | Full Syllabus |
| **Chemistry:** | Full Syllabus |
| **Mathematics:** | Full Syllabus |

**General Instructions**:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 75 questions. The maximum marks are **300**.
4. There are three Sections in the question paper, Section I, II & III consisting of Section-I **(Physics)**, Section-II **(Chemistry)**, Section-III **(Mathematics)** and having **25** questions in each part in which first **20** questions are of Objective Type and Last **5** questions are integers type and all **25** questions are **compulsory.**
5. There is only one correct response for each question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

**OMR Instructions:**

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

**Name of the Student (In CAPITALS) :** _____

**Roll Number:** _____

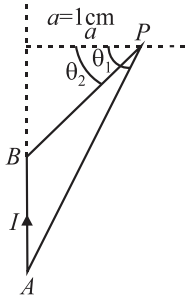
**OMR Bar Code Number:** _____

**Candidate's Signature:** _____ **Invigilator's Signature** _____

## SECTION-I (PHYSICS)

***Single Correct Type Questions***

**1.** The straight wire *AB* carries a current $I$ = 1A. The ends of the wire subtend angles $\theta_1 = 53°$ and $\theta_2 = 37°$ respectively at the point *P* as shown in figure. The magnetic field at the point *P* is




(1) 1 μT (2) 2 μT
(3) 3 μT (4) 4 μT

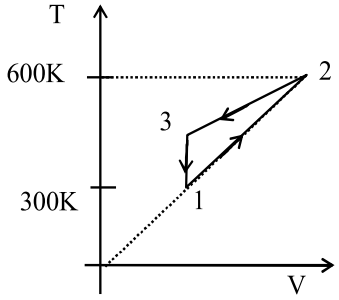
**2.** What should be the approximate velocity of a proton so that its momentum becomes equal to that of a photon of wavelength 5200 Å? $\left(\text{Given}\frac{m_P}{m_e}=1800\right)$

(1) 0.99 m/s (2) 0.11 m/s
(3) 0.77 m/s (4) 0.33 m/s

**3.** A moving hydrogen atom makes a head-on collision with a stationary hydrogen atom. Before collision, both atoms are in ground state and after collision they move together. The minimum value of the kinetic energy of the moving hydrogen atom, such that both of the atoms reaches one of the excitation state is

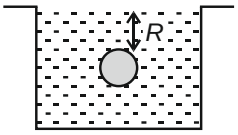
(1) 20.4 eV (2) 10.2 eV
(3) 54.4 eV (4) 40.8 eV

**4.** Two moles of an ideal diatomic gas is taken through a cyclic process 1-2-3-1 as shown in the *T*-*V* diagram. If the net heat rejected in the cyclic process is 300 J, then the ratio of the net work done by the gas in the cyclic process to the heat supplied (Assume universal gas constant $R$= 8.3 J/mol K)



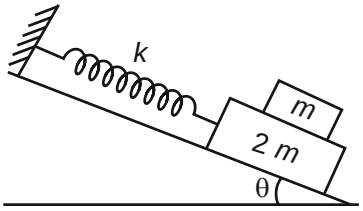

(1) $\frac{1}{5R}$ (2) $-\frac{1}{3R}$
(3) $-\frac{1}{7R}$ (4) $-\frac{3}{5R}$

**5.** A sphere of radius $R$ is submerged in a liquid of density ρ. The force exerted by liquid on upper half of sphere at the position shown in figure, is (assume atmospheric pressure to be zero)




(1) $\frac{3}{2}\pi R^3\rho g$
(2) $2\pi R^3\rho g$
(3) $\frac{1}{3}\pi R^3\rho g$
(4) $\frac{4}{3}\pi R^3\rho g$

**6.** The coefficient of friction between block of mass $m$ and $2m$ is $\mu = 2\tan\theta$. There is no friction between block of mass $2m$ and inclined plane. The elongation in spring at mean position is




(1) $g\sin\theta\sqrt{\frac{k}{m}}$ (2) $\frac{mg\sin\theta}{k}$
(3) $\frac{3mg\sin\theta}{k}$ (4) $\frac{2mg\sin\theta}{k}$

**7.** The energy released by the fission of a single uranium nucleus is 200 MeV. The number of fissions of uranium nucleus per second required to produce 16 MW of power is
(Assume efficiency of the reactor is 50%)

(1) $2\times10^{18}$ (2) $2.5\times10^{18}$
(3) $5\times10^{18}$ (4) $10^{18}$

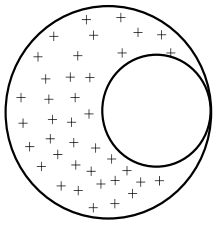
**8.** A capacitor of capacitance $C$ is charged with charge $Q$ and then connected to another uncharged capacitor of capacitance '$c$' and subsequently disconnected. The latter is completely discharged by short circuiting its plates and then again connected to the first capacitor. The process is repeated $n$ times. Then the charge left on the first capacitor is given by

(1) $\left(\frac{C}{C+c}\right)^n Q$ (2) $\left[\frac{C+c}{C}\right]^n Q$
(3) $\left[\frac{C+c}{C}\right]^{2n} Q$ (4) $\left[\frac{C}{C+c}\right]^{2n} Q$

**9.** A solid non-conducting sphere having uniform charge density ρ and radius $R$ is shown in figure.

A spherical cavity of radius $\frac{R}{2}$ is hollowed out.

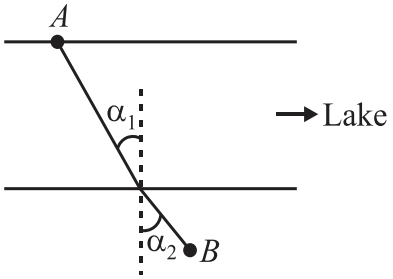
The potential at centre of bigger sphere is (Assuming zero potential at infinity)



(1) $\frac{11R^2\rho}{24\varepsilon_0}$ (2) $\frac{5R^2\rho}{12\varepsilon_0}$

(3) $\frac{7\rho R^2}{12\varepsilon_0}$ (4) $\frac{3R^2\rho}{2\varepsilon_0}$

**10.** From a point $A$ on the banks of a lake with still water a man must get to a point $B$ on the opposite side of the lake. The man uses a boat to travel across the lake and then walk on the land to reach point $B$. His rowing and walking speeds are $v_1 = 6\,\text{m/s}$ and $v_2 = 2\,\text{m/s}$ respectively. If man moves from point $A$ to $B$ in minimum possible time, then value of $\frac{\sin\alpha_1}{\sin\alpha_2}$ is equal to




(1) $\sqrt{3}$ (2) $\frac{1}{\sqrt{3}}$

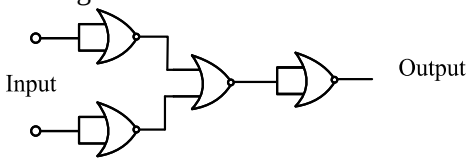
(3) 3 (4) $\frac{1}{3}$

**11.** When a capillary is dipped vertically inside a liquid, the liquid rises up to a height $h$. In the same capillary, the same liquid is filled in horizontal position and then kept in vertical position with both ends in air. The maximum length of liquid column which can be kept in vertical position is

(1) $\frac{h}{2}$ (2) $h$

(3) $2h$ (4) $\frac{3h}{2}$

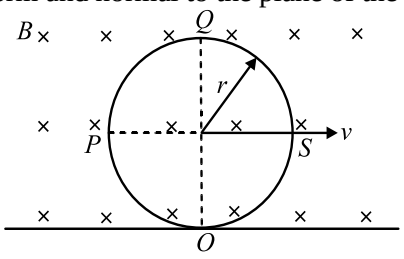
**12.** From the combination of four logic gates following circuit is devised




(1) OR (2) AND

(3) NAND (4) XOR

**13.** A conducting ring of radius $r$ and resistance $R$ rolls without slipping on a horizontal surface with constant velocity $v$. The magnetic field $B$ is uniform and normal to the plane of the loop then



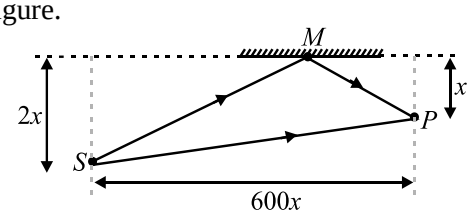

(1) The induced emf between $O$ and $Q$ is $2Bvr$

(2) An induced current $I = \frac{2Bvr}{R}$ flows in the clockwise direction

(3) An induced current $I = \frac{2Bvr}{R}$ flows in the anticlockwise direction

(4) The induced emf between $O$ and $Q$ is $Bvr$

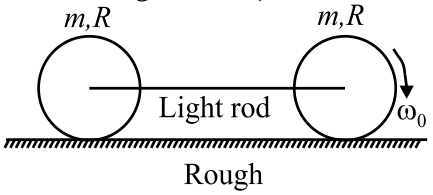
**14.** The point $S$ is a monochromatic source of light emitting light of wavelength λ. At the point $P$ interference takes place between two light rays, one directly coming from $S$ and other after reflection from the mirror. At $P$ maxima is formed. Find minimum value of $x$, shown in figure.




(1) 60λ (2) 65λ

(3) 75λ (4) 85λ

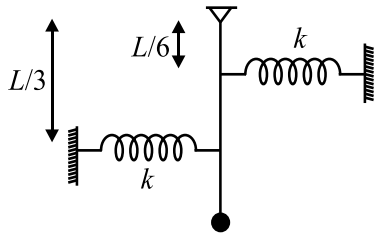
**15.** The assembly of two identical disk ($m$, $R$) connected by light rigid rod is kept on a sufficient rough horizontal surface. Front disk is given an initial angular velocity $\omega_0$ = 12 rad/s. Total impulse (in SI units) provided by all the forces on rear disk till the time it start pure rolling motion is (Given $m$ = 2 kg, $R$ = 1 m)




(1) 1 (2) 2

(3) 3 (4) 4

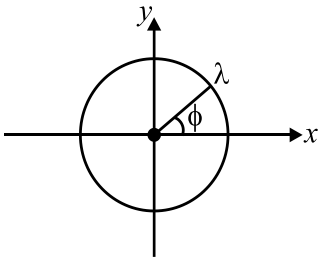
**16.** A uniform rod of mass $2m$ and length $L$ is hinged at one end and carries a particle of mass $m$ at the other end. Two springs each of force constant $k$ are installed at distances as shown. The whole arrangement rests on a smooth horizontal surface. The frequency of small oscillations will be




(1) $\frac{1}{2\pi}\sqrt{\frac{k}{m}}$ (2) $\frac{1}{2\pi}\sqrt{\frac{k}{3m}}$

(3) $\frac{1}{4\pi}\sqrt{\frac{k}{3m}}$ (4) $\frac{1}{2\pi}\sqrt{\frac{3k}{m}}$

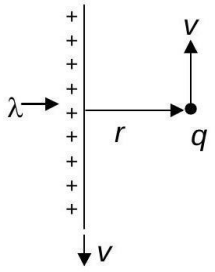
**17.** A thin non-conducting ring of radius $a$ has a linear charge density $\lambda = \lambda_0 \sin\phi$. A uniform electric field $E_0\hat{i} + E_0\hat{j}$ is present in the region. Magnitude of torque acting on the ring is




(1) Zero (2) $2E_0\pi a^2\lambda_0$

(3) $E_0\pi a^2\lambda_0$ (4) $E_0\frac{\pi a^2}{2}\lambda_0$

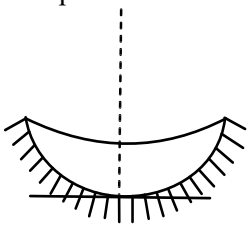
**18.** A long wire having linear charge density $\lambda$ moving with constant velocity $v$ along its length. A point charge moving with same speed in opposite direction and at that instant it is $r$ distance from the wire. The net force acting on the charge is given by




(1) $\frac{\lambda q}{2\pi r}\left[\frac{1}{\varepsilon_0} + v^2\mu_0\right]$ (2) $\frac{\lambda q}{2\pi r}\left[\frac{1}{\varepsilon_0} - \mu_0 v^2\right]$

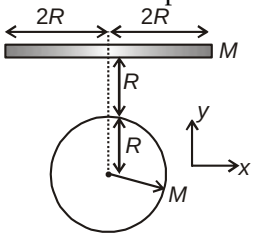
(3) $\frac{\lambda q}{2\pi r}\sqrt{\left(\frac{1}{\varepsilon_0}\right)^2 + v^4\mu_0^2}$ (4) zero

**19.** The convex surface of a thin concavo-convex lens of glass of refractive index 1.5 has a radius of curvature 20 cm. The concave surface has a radius of curvature 60 cm. The convex side is silvered and placed on a horizontal surface. At what distance from horizontal surface should a pin be placed on optical axis so that image is formed at same place.



(1) 5 cm (2) 10 cm
(3) 20 cm (4) 15 cm

**20.** Consider the arrangement shown in figure. A thin rod (mass $M$ and length $4R$) and a sphere of equal mass $M$ and radius $R$ are located as shown in figure. The centre of the sphere being at a distance $2R$ from the centre of the rod on its perpendicular bisector. Assume that the system of rod and sphere is in gravity free space. Find the force acting on the rod due to sphere.



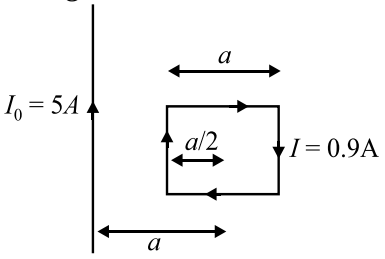

(1) $\frac{GM^2}{4\sqrt{2}R^2}\left(\hat{j}\right)$ (2) $\frac{GM^2}{4R^2}\left(-\hat{j}\right)$

(3) $\frac{GM^2}{4\sqrt{2}R^2}\left(-\hat{j}\right)$ (4) $\frac{GM^2}{2\sqrt{2}R^2}\left(-\hat{j}\right)$

***Integer Type Questions***

**21.** Light coming from a discharge tube filled with singly ionised helium atom is incident on the cathode of photoelectric cell, with work function 5 eV. The minimum value of cathode voltage (with respect to the anode) to make photo current zero is $x$ volt. Find the value of $10x$.

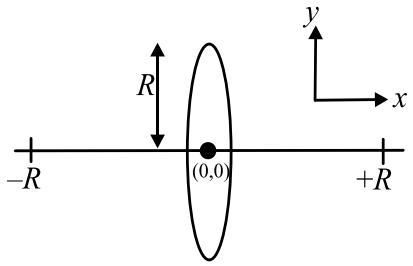
**22.** A square frame of side length $a$ = 8.0 cm carries a current $I$ = 0.9 A and is located in same plane as a long straight wire carrying current $I_0$ = 5 A, as shown in figure.




The magnetic force of interaction between the wire and the frame is $k \times 10^{-7}$ N. Find the value of $k$.

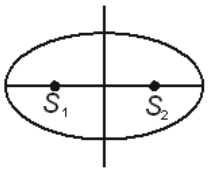
**23.** A circular wire loop of radius $R$ carrying current $I$ as shown in figure. Axis of the loop is along '$x$' axis and center of the loop lies at origin. The line integral of resulting magnetic field (by circular loop) along $x$-axis from $x = -R$ to $x = +R$

$\int_{-R}^{+R} B.dx = \frac{\mu_0 I}{\sqrt{k}}$. Find the value of $k$.




**24.** A rod of length $L$ kept on a smooth horizontal surface is pulled along its length by a force $F$. The area of cross section is $A$ and Young's modulus is $Y$. If the elastic potential energy stored in the rod is $U_0$. Find the value of $\frac{F^2 L}{U_0 AY}$.

**25.** $S_1$ and $S_2$ are two coherent sources of sound vibrating in same phase located at two focci of an ellipse of eccentricity $e = 0.3$ and length of major axis $10\lambda$ where $\lambda$ is wavelength of sound emitted by them. The number of maxima located on the ellipse will be




## SECTION-II (CHEMISTRY)

***Single Correct Type Questions***

**26.** The electron are identified by the quantum numbers 'n' and '$\ell$'

I. n = 4, $\ell$ = 1 II. n = 4, $\ell$ = 0

III. n = 3, $\ell$ = 2 IV. n = 3, $\ell$ = 1

Can be placed in order of increasing energy as

(1) III < IV < II < I
(2) IV < II < III < I
(3) II < IV < I < III
(4) I < III < II < IV

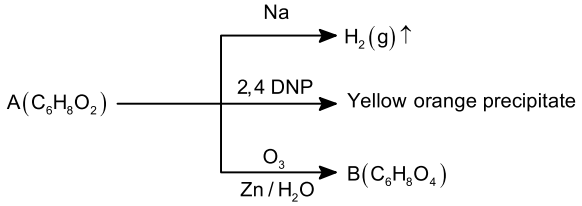
**27.** pH of a saturated solution of $Ba(OH)_2$ is 12. Hence, the $K_{sp}$ of the $Ba(OH)_2$ is

(1) $5 \times 10^{-7}$ (2) $5 \times 10^{-4}$
(3) $10^{-6}$ (4) $4 \times 10^{-5}$

**28.** A gas occupies 2 litre volume at STP. It is provided 300 Joule heat so that its volume becomes 2.5 litre at 1 atm. Change in internal energy would be nearly

(1) 239 Joule (2) 205 Joule
(3) 249.37 Joule (4) 220.37 Joule

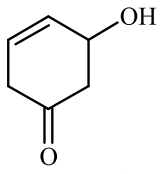
**29.** The compound A gives the following reactions

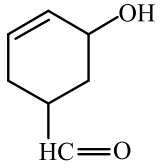



The structure of A can be

(1) $CH_2=CH-(CH_2)_2-\underset{\underset{O}{\|}}{C}-CH_2OH$

(2) $H-\underset{\underset{O}{\|}}{C}-(CH_2)_2-CH=CH-COOH$

(3) 

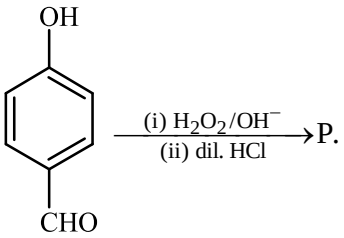

(4) 


**30.** If $E^{\circ}_{BrO_3^-/BrO^-} = 0.54$ volt and

$E^{\circ}_{BrO^-/\frac{1}{2}Br_2} = 0.41$ volt

Then, find out the value of $E^{\circ}_{BrO_3^-/\frac{1}{2}Br_2}$

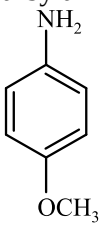
(1) 0.35 volt (2) 0.52 volt
(3) 0.71 volt (4) 0.66 volt

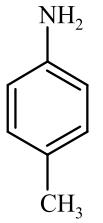
**31.** 

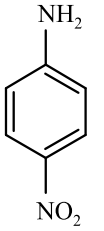

Which of the following statement is wrong regarding the product?

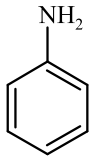
(1) The product gives violet colouration with $FeCl_3$
(2) The product participates in Reimer Tiemann reaction
(3) The product reacts with aq. $NaHCO_3$
(4) The product reacts with aq. NaOH

**32.** Which of the following compounds responds in the carbylamine test in the fastest rate?

(1) 

(2) 

(3) 

(4) 

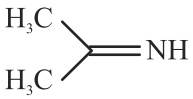
**33.** The increasing order of atomic radii of the following Group 13 elements is

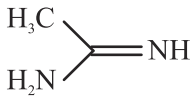
(1) Al < Ga < In < Tl
(2) Ga < Al < In < Tl
(3) Al < In < Ga < Tl
(4) Al < Ga < Tl < In

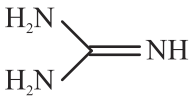
**34.** The correct order of the oxidation states of nitrogen in NO, $N_2O$, $NO_2$ and $N_2O_3$ is

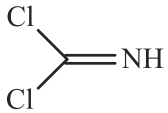
(1) $NO_2 < NO < N_2O_3 < N_2O$
(2) $N_2O < NO < N_2O_3 < NO_2$
(3) $NO_2 < N_2O_3 < NO < N_2O$
(4) $N_2O < N_2O_3 < NO < NO_2$

**35.** Which among the following compounds is most basic?

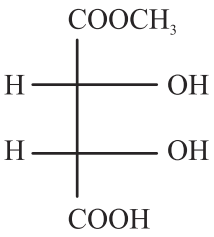
(1) 

(2) 

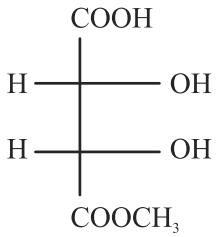
(3) 

(4) 

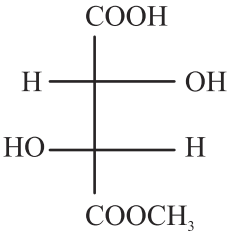
**36.** The correct statement about the compound A, B and C is



(A)



(B)



(C)

(1) 'A' and 'B' are identical
(2) 'A' and 'B' are diastereomers
(3) 'A' and 'C' are enantiomers
(4) 'A' and 'B' are enantiomers

**37.** Observer the following molecules

$PCl_5$ , $BrF_5$ ,$ClF_5$ , $PF_5$ ,$ClF_3$ , $XeF_4$ , $XeF_2$ , $IF_5$

The number of molecules having square pyramidal geometry from the above is:

(1) 4
(2) 5
(3) 3
(4) 6

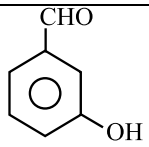
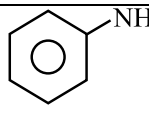
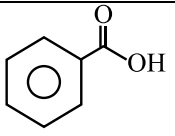
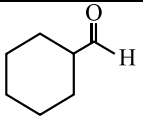
**38.** The colour of potassium dichromate is due to:

(1) d-d transition
(2) transition in $K^+$ ion
(3) ligand-to-metal charge transfer
(4) metal to ligand charge transfer

**39.** Incorrect statement for metal carbonyl $[M(CO)_6]$ is:

(1) σ bonding electrons are donated by CO group to the metal.
(2) dπ – pπ back bonding occurs from the filled metal orbitals.
(3) dπ – pπ back bonding makes M-CO bond stronger.
(4) C–O bond length is greater in CO gas as compared to C–O bond length in given complex.

**40.** Match the following

| List-I (Compound) | | List-II (Identifying test) | |
|---|---|---|---|
| A |  | P | Carbylamine test |
| B |  | Q | Fehling's solution test |
| C |  | R | $NaHCO_3$ test |
| D |  | S | Neutral $FeCl_3$ test |

(1) A→S; B→P; C→Q; D→R
(2) A→S; B→P; C→R; D→Q
(3) A→Q; B→S; C→R; D→P
(4) A→Q; B→P; C→R; D→S

**41.** The following data were obtained during the first order thermal decomposition of $SO_2Cl_2$ at a constant volume:

$$SO_2Cl_2(g) \rightarrow SO_2(g) + Cl_2(g)$$

| Exp | Time (sec) | Total pressure (atm) |
|---|---|---|
| 1 | 0 | 0.5 |
| 2 | 100 | 0.6 |

Find out rate of reaction (atm $sec^{-1}$) when total pressure is 0.65 atm. (Initially only $SO_2Cl_2$ is taken) (log2 = 0.3, log3 = 0.48, log4 = 0.6, log 5 = 0.7)

(1) $1.5 \times 10^{-3}$ (2) $8.06 \times 10^{-4}$

(3) $2.3 \times 10^{-3}$ (4) $4.03 \times 10^{-4}$

**42.** Which one is incorrect statement amongs the following?

(1) $PH_5$, $SCl_6$ and $FCl_3$ do not exist.

(2) 12 P–O bonds are present in $P_4O_6$ molecule.

(3) Cation of $PCl_5(s)$ has tetrahedral shape.

(4) Bond angle in $SiH_4$ is less than that in $CH_4$.

**43.** Which is the correct match for all?

| List-I | | List-II | |
|---|---|---|---|
| A | Sucrose | P | β-(1,4)-Glycosidic linkage |
| B | Maltose | Q | α, β-(1,2)-Glycosidic linkage |
| C | Starch | R | natural biopolymer |
| D | Cellulose | S | Silver mirror with tollen's reagent |

(1) (A-S); (B-P); (C-R); (D-Q)

(2) (A-Q); (B-S); (C-P); (D-R)

(3) (A-Q); (B-S); (C-R); (D-P)

(4) (A-R); (B-S); (C-Q); (D-P)

**44.** A sample of 3 gram (92% pure) of silver carbonate on being strongly heated yields a residue and gaseous products. Find the volume of evolved gases at STP. (Given, molar mass of Ag = 108 g/mol)

(1) 112 mL (2) 22 mL

(3) 336 mL (4) 33.6 mL

**45.** Ratio of $\Delta T_b/K_b$ for 6% (W/V) $AB_2$ is 1 mol/kg and also for 9% (W/V) $A_2B$ it is 1 mol/kg ($AB_2$ and $A_2B$ both are non-elecrolytes). Hence, atomic masses (in amu) of A and B are respectively : (assume molality = molarity)

(1) 60, 90 (2) 40, 40

(3) 40, 10 (4) 10, 40

***Integer Type Questions***

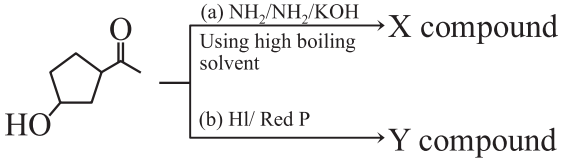
**46.** At 1000 K, a sample of pure $NO_2$ decomposed as

$2NO_2(g) \rightleftharpoons 2NO(g) + O_2(g)$

The equilibrium constant $K_P$ = 156.25 atm. If partial pressure of $O_2$ is 0.25 atm at equilibrium, and the partial pressure of $NO_2$ at equilibrium is x atm, then the value of 100 x is:

**47.** The sum of group and period number of an element with atomic number 70 is:

**48.**




Difference in molecular weight of compound X and Y is equal to Z g/mol, then numerically $\sqrt{Z}$ is equal to

(Given, molar mass in g/mol of C-12, H-1, O-16)

**49.** How many of the following has hydrogen bonding

(a) $NH_3$ (b) $CH_4$

(c) $H_2O$ (d) HI

(e) HF (f) HCOOH

(g) $B(OH)_3$ (h) $CH_3COOH$

**50.** Aniline and N-methylaniline can be distinguished by using, how many of the following tests?

(I) Tollen's reagent (II) Hinsberg reagent

(III) Carbylamine test (IV) Mulliken Test

(V) Azo dye test (VI) Bayer's test

(VII) Lucas test (VIII) Hofmann mustard oil test

## SECTION-III (MATHEMATICS)

***Single Correct Type Questions***

**51.** The number of real solution of the equation $\sqrt{x^2-4x+3}+\sqrt{x^2-9}=\sqrt{4x^2-14x+6}$, is:

(1) 0

(2) 1

(3) 3

(4) 2

**52.** Let $f: R \rightarrow [2, \infty)$ be a function defined as $f(x) = x^2 - 12ax + 15 - 2a + 36a^2$. If $f(x)$ is surjective on $R$, then the value of $a$ is equal to

(1) $\frac{9}{2}$ (2) $\frac{11}{2}$

(3) $\frac{13}{2}$ (4) $\frac{15}{2}$

**53.** Let two sides of a rectangle of area 20 sq. units are along lines $x - y = 0$ and $x + y = 2$, then the locus of the point of intersection of diagonals is a

(1) pair of ellipses

(2) pair of straight lines

(3) pair of hyperbolas having eccentricity 2 and $\frac{2}{\sqrt{3}}$

(4) pair of hyperbolas each having eccentricity $\sqrt{2}$

**54.** Let $A=\begin{bmatrix}2 & 0 & 7\\ 0 & 1 & 0\\ 1 & -2 & 1\end{bmatrix}$ and $B=\begin{bmatrix}-\alpha & 14\alpha & 7\alpha\\ 0 & 1 & 0\\ \alpha & -4\alpha & -2\alpha\end{bmatrix}$. If $AB = I$, where $I$ is an identity matrix of order 3 then trace $B$ has value equal to

(1) 0 (2) $\frac{2}{5}$

(3) $\frac{1}{5}$ (4) 5

**55.** Consider the points $A$, $B$, $C$ and $D$ with position vector $7\hat{i}-4j+7k, \hat{i}-6j+10k, -\hat{i}-3j+4k$ and $5\hat{i}-\hat{j}+\hat{k}$ respectively, then $ABCD$ is a

(1) square

(2) rhombus

(3) rectangle

(4) parallelogram but not a rhombus

**56.** Let $A(1, 4)$ and $B(1, -5)$ be two points. Let $P$ be a point on the circle $(x - 1)^2 + (y - 1)^2 = 1$ such that $(PA)^2 + (PB)^2$ have maximum value, then the points $P$, $A$ and $B$ lie on

(1) a straight line

(2) a equilateral triangle

(3) an isosceles triangle

(4) None of these

**57.** If $x = 1$ is a critical point of the function $f(x) = (3x^2 + ax - 2 - a)e^x$, then:

(1) $x = 1$ and $x=-\frac{2}{3}$ are local minimum of $f$

(2) $x = 1$ and $x=-\frac{2}{3}$ are local maximum of $f$

(3) $x = 1$ is a local maximum and $x=-\frac{2}{3}$ is a local minimum of $f$

(4) $x = 1$ is a local minimum and $x=-\frac{2}{3}$ is a local maximum of $f$

**58.** If $\cos 5\theta = 5\cos\theta - 20\cos^3\theta + a\cos^5\theta + b$, then the value of $a + b$ is equal to

(1) 20 (2) 16

(3) –16 (4) 15

**59.** If $f: R \to R$ is a function defined by $f(x) = [x - 1]\cos\left(\frac{2x-1}{2}\right)\pi$, where [.] denotes the greatest integer function, then $f$ is:

(1) Discontinuous at all integral values of $x$ except at $x = 1$

(2) Continuous only at $x = 1$

(3) Continuous for every real $x$

(4) Discontinuous only at $x = 1$

**60.** If $f(x)=\prod_{i=1}^{i=3}(x-a_i)+\sum_{i=1}^{3}a_i-3x$, where $a_i < a_{i+1}$, then $f(x) = 0$ has

(1) Only one real root

(2) Three real roots of which two of them are equal

(3) Three distinct real roots

(4) Three equal roots

**61.** Let $S$ be the set of all values of $\lambda$, for which the shortest distance between the lines $\frac{x-\lambda}{0}=\frac{y-3}{4}=\frac{z+6}{1}$ and $\frac{x+\lambda}{3}=\frac{y}{-4}=\frac{z-6}{0}$ is 13. Then $8\left|\sum_{\lambda\in S}\lambda\right|$ is equal to

(1) 302 (2) 306

(3) 304 (4) 308

**62.** Let $a$, $b$, $c$ be three real numbers such that $a\in\left[0,\frac{\pi}{2}\right], b\in R-\{0\}$ and $c \in (-\infty, \infty)$. If $\left|(\sin a+\cos a)-\sqrt{2}\right|+\left|b+\frac{1}{b}-2\right|+|c|\le 0$, then find the value of $\begin{vmatrix}\sin(4a) & \cos^2 a & 1\\ b+2 & b^3 & \frac{1}{b}\\ \ln(c+1) & c+5 & c^3+2\end{vmatrix}$

(1) 10 (2) 11

(3) 12 (4) 9

**63.** The number of rational terms in the expansion of $\left(\sqrt{2}+\sqrt[3]{3}+\sqrt[6]{5}\right)^{10}$ is equal to

(1) 0 (2) 1

(3) 2 (4) 3

**64.** Let f : R $\to$ R be a differentiable function such that $f\left(\frac{\pi}{4}\right)=\sqrt{2}, f\left(\frac{\pi}{2}\right)=0$ and $f'\left(\frac{\pi}{2}\right)=1$ and let $g(x)=\int_{x}^{\pi/4}\left(f'(t)\sec t+\tan t\sec t f(t)\right)dt$ for $x\in\left[\frac{\pi}{4},\frac{\pi}{2}\right)$. Then $\lim\limits_{x\to\left(\frac{\pi}{2}\right)^{-}} g(x)$ is equal to

(1) 2 (2) 3
(3) 4 (4) –3

**65.** If $2x^y + 3y^x = 20$, then $\frac{dy}{dx}$ at (2, 2) is equal to:

(1) $-\left(\frac{3+\log_e 8}{2+\log_e 4}\right)$ (2) $-\left(\frac{2+\log_e 8}{3+\log_e 4}\right)$
(3) $-\left(\frac{3+\log_e 4}{2+\log_e 4}\right)$ (4) $-\left(\frac{3+\log_e 16}{4+\log_e 8}\right)$

**66.** The value of the integral $\int\frac{(1-\cos\theta)^{2/7}}{(1+\cos\theta)^{9/7}}d\theta$ is

(1) $\frac{7}{11}\left(\tan\frac{\theta}{2}\right)^{\frac{11}{7}}+c$ (2) $\frac{7}{11}\left(\cos\frac{\theta}{2}\right)^{\frac{11}{7}}+c$
(3) $\frac{7}{11}\left(\sin\frac{\theta}{2}\right)^{\frac{11}{7}}+c$ (4) $\frac{7}{11}\left(\cot\frac{\theta}{2}\right)^{\frac{11}{7}}+c$

**67.** Let a dice be rolled *n* times. Let the probability of getting odd numbers seven times be equal to the probability of getting odd numbers nine times. If the probability of getting even numbers twice is $\frac{k}{2^{15}}$, then *k* is equal to

(1) 60 (2) 30
(3) 90 (4) 15

**68.** If *x*, *y*, *z* are positive and *x* + *y* + *z* = 1, then the minimum value of $\left(\frac{1}{x}-1\right)\left(\frac{1}{y}-1\right)\left(\frac{1}{z}-1\right)$ is

(1) 12
(2) 4
(3) 8
(4) 9

**69.** Let *y* be the solution of the differential equation $(1-x^2)dy=\left(xy+\left(x^3+2\right)\sqrt{1-x^2}\right)dx$, $-1<x<1$ and $y(0)=0$. If $\int_{-\frac{1}{2}}^{\frac{1}{2}}\sqrt{1-x^2}\,y(x)dx=k$, then 80 *k* is equal to

(1) 0.125 (2) 0.25
(3) 0.5 (4) 1

**70.** Solution set of the in equality

$(\log_{10}x)^2-3(\log_{10}x)(\log_{10}(x-2))$

$+2\left(\log_{10}(x-2)\right)^2<0$, is:

(1) (0, 4)
(2) $(-\infty, 1)$
(3) $(4, \infty)$
(4) (2, 4)

***Integer Type Questions***

**71.** Let $A = \{1, 2, 3, 4\}$. If the number of different ordered pairs $(B, C)$ that can be formed such that $B\subseteq A$, $C\subseteq A$ and $B\cap C=\phi$ is n then find $\sqrt{n}$.

**72.** Let $z = 1 + i$ and $z_1=\frac{1+i\bar{z}}{\bar{z}(1-z)+\frac{1}{z}}$. Then $\frac{12}{\pi}\arg(z_1)$ is equal to_____.

**73.** If $y=\sum_{k=1}^{6}k\cos^{-1}\left\{\frac{3}{5}\cos kx-\frac{4}{5}\sin kx\right\}$, then $\frac{dy}{dx}$ at $x = 0$ is

**74.** Let $S_n$ and $\Delta_n$ be the sum of first n terms of 2 A.P's whose $r^{th}$ terms are $T_r$ and $t_r$ respectively. If $\frac{S_n}{\Delta_n}=\frac{2n+5}{3n+2}$ then $\frac{T_{11}}{t_{10}}=\frac{m_1}{m_2}$ where $m_1$ and $m_2$ are coprime. Find the value of $\frac{m_2-m_1}{2}$.

**75.** Let $f(x)$ be a polynominal satisfying $\lim\limits_{x\to\infty}\frac{x^2f(x)}{2x^5+3}=6$, also $f(1) = 3$, $f(3) = 7$ and $f(5) = 11$, then the value of $\left(\frac{f(6)+5f(4)}{29}\right)$ is



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